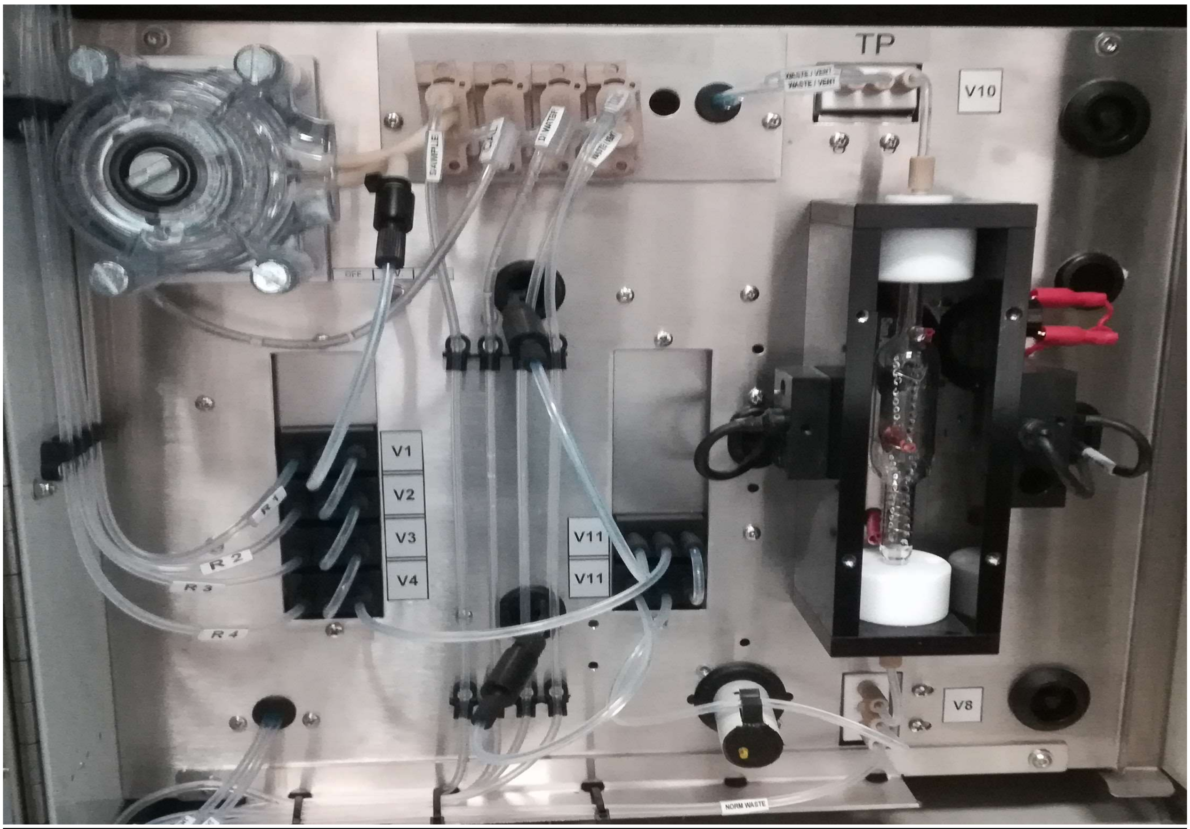
Operating Manual for the µMAC TP

(Vs. 1.1)

for the analysis of total phosphorus in water and waste water
Addendum to the Micromac operating manual.



General description

The µMAC TP is an analyser for the automated analysis of the total phosphorus in water, using the microLFR HT principle, in which the micro LFR technique is used together with an high temperature digester.
The analysis is performed automatically with two steps of digestion with acid and persulfate, first in UV then in a high temperature digester for the recovery of both organic and inorganic compounds; the measurement of the orthophosphate is done by molybdenum blue method.

For TP, the sample is pumped and dosed inside the digestion tube, an eventual dilution is done depending on the range, then solutions of acid and oxidant are dispensed: the mixture is then put into a UV digester; after a set time of UV digestion, the mixture if put back in the heating digester and a temperature digestion is started; at the end of the heating the mixture is cooled down and the reagents for PO4 measurement are added, the colour is produced inside the digester that is also the flowcell for the reading of PO4 and the OD is measured; the final product is then wasted and the digester and UV and other components of the hydraulic are washed. The digestion times can be selected depending on the matrix.

The digester is left with DI water.

The typical samples are drinking water, surface water, waste water and industrial water.

The sample is aspirated and dosed by the system internal pump, and the reagents are dosed by the activation of the same pump: one single pump is used.

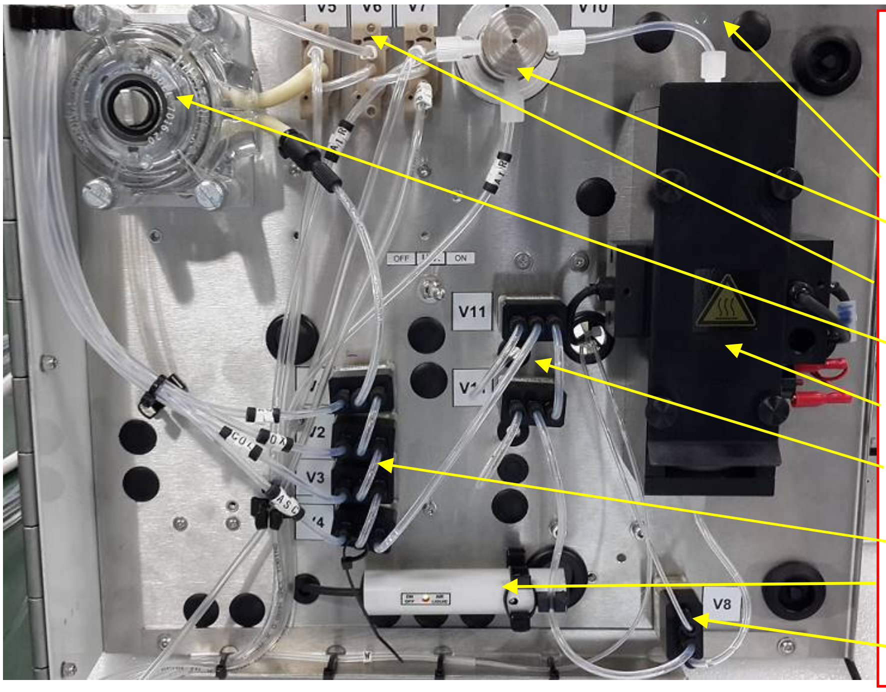
The calibration of the system is performed aspirating two solutions, one at zero concentration (DI water) and the other using a calibrant, made of phosphate for TP. The calibrant concentration is within the linearity range of the system.

The storage of the data in the internal memory is practically unlimited when used with touch screen, while for keypad and keyboard the memory is of 2000 results (rolling memory), and the last result can be set available for 4-20 mA output for the methods.
The reagent compartment allows the storage of reagents for about one litre acid reagent and one litre of oxidant, and also 1000 mL for each of the calibrant solutions.

For particularly hot environment, a special reagent compartment fitted with Peltier cooling device is available.

The system is operating at 12 VDC. Note that the power of the heating bath is separated from the power of the system and can be set at different voltage, normally it is set at 13 VDC for the electronic and 17 VDC for the heating bath, powered through a rele.

General overview of the manifold of the system, with some components described. Note that this picture represents a kit mounted outside the cabinet.



Description of main components: refer to the flowchart at end of this addendum for further information

Reagent compartment (above manif.)

Digester top valve

Sample and calibrators valves V5-7

Peristaltic pump

Temperature digester and colorimeter

UV digester bypass valves V11

Reagent valves V1-4

Sample detector sensor

Waste valve V8

Principle of analytical cycle for TP

The following steps describe the working principle applied for the TP method:

1) *The digester is emptied in back flush through the vent valve*
2) *The sample is aspirated and primed through the V8*
3) *The sample is dosed on time base in the digester (depending on range)*
4) *The circuit is emptied*
5) *The eventual dilution water is primed and then dosed same way of sample on time base*
6) *The acid and the oxidant are dosed in the digester, on time base (2+2 seconds)*
7) *The mixture is moved to the UV digester and the first digestion is started*
8) *After the set time, the mixture is moved to the digester*
9) *The digestion at boiling point starts: a minimum of some minutes is set, the further digestion time can be set accordingly to manual method, normally it is 10 minutes; digestion is done under pressure with both top and bottom valves closed*
10) *The digestion ends, a cooling phase starts to bring solution below 55°C*
11) *The OD start is measured to compensate the colour of sample after dilution and digestion*
12) *The addition of the colour reagent and reducing agent for the PO4 measurement is done, the colour reaction starts*
13) *At end of reaction time, the final OD is taken and the system makes the calculation*
14) *The digester is emptied*
15) *DI water is aspirated several times to wash the digester, the UV digester all the hydraulic parts, comprehending the top valve for pressurized digestion*
16) *A final amount of DI water is left inside the digester, to ensure a better cleaning*

If the monitoring function is active, the system will wait till the expiry of the set interval time and then starts a new cycle, otherwise the system will stay in ready mode, and an analysis can be started at any time; during the monitoring interval, a complete cycle may be started at any time by the keyboard, the external contact or via the serial port, if the system is not in busy.

System preparation – start up quick reference guide

Place the system as described on the manual on a convenient stable position, and connect the power cables as reported on the connector inside the electronic compartment; do not start the power supply yet; make sure that the power supply is with earth (ground) connection, to ensure stability of power supply and safety of the system.
Prepare the reagents as described further on in this manual, and put the reagents inside proper containers inside the reagent compartment, put the wastes lines in a waste tank: ensure that no waste lines are getting in touch with the liquid, they must be just under the cap for free aspiration of vent during the system functioning (systems with this need are normally shipped with a cap already prepared for this purpose as example).

Once all the above settings have been done, follow next procedure to make system start up:

1) the system is shipped empty to avoid freezing during long haul flights: run a wash cycle to ensure that all connections are safe and there is no leak at any side of the manifold, this cycle will also put DI water inside the digester
2) put the reagents straws as described in the table above, AC is for acid, OX is for oxidant, COL is for colour reagent for TP, ASC is ascorbic acid for TP, C is for calibrant, H is for water and S is for sample (the last two are normally outside the system)
3) Start a PRIME cycle and observe that all the reagents are properly pumped (sequence is: acid, oxidant, colour, ascorbic, then washes) the reagents go straight into waste line through V8
4) Start a Blank cycle and check its value with the stored value: in this system the blank represents water with reagents, with normal digestion time: the values must be close to zero, check against the stored values
5) Start a calibration with 0.5 ppm P-PO4 (or the value set as calibrant concentration) and check the values with the stored ones: in this sequences the digestion timings are standard (note: if the value exceeds a preset tolerance, the message "calibration error" appears: repeat the calibration and eventually, if the error persists, save it in the calibration menu with [update last cal])
6) Start an analysis cycle running a 50 % calibrant (dilute the calibrant 50/50 with the DI water) to check the system performances, reading should be in the range of 50 % pf the calibrant ±10%.
7) System is now ready to work.

System preparation for shipment – storage – shut down

Before a long period of no operation of the system (more than two weeks time), it is advisable to leave the system clean, with water inside; before a shipment of the system it is advisable to clean the system and then empty its hydraulic parts, to leave it ready for next usage without risks of dirtying or freezing of water inside components.
To perform this cleaning, follow next procedure to make a shut down:

1) Put all the reagent straws, sample and calibrant lines in air and start a prime cycle, to empty reagent, sample and calibrant lines
2) Put all the reagent, calibrant and sample straws into a container with DI water and start a prime, to wash all the valves and lines
3) Repeat the prime twice or until no residual colour is visible in the lines
*4) The system is now ready for **storage**, with water inside, valves cleaned and lines without any calibrant or sample*
*5) If the system has to be shipped, repeat the prime with all lines in air and start several primes to empty all hydraulic manifold: the system is now ready for **shipment***
6) Switch off the power to the system and disconnect the cablings from the electronic compartment, insert all the hydraulics inside the system from the bottom, remove all bottles from inside the system and ensure that the doors are all safely locked before moving the system; it is also suggestible to block on site the reagent plastic plate during shipment with some tape.

The meaning of washing the system hydraulic is to avoid crystallization inside the valves of the reagent salts that needs eventually a hot water wash with manual valve excitement to be cleaned.

The meaning of emptying the system hydraulic is to avoid icing of the manifold with unpredictable consequences during the low temperatures reached during long haul shipment.

Operating notes

- *During the automatic recalibration, on the proper field in the [Monitor] menu set both the blank and calibrant to be performed (B+C shown on the setting of the Cal Type). Please note that the tolerances for both blank and calibrant ODs will be applied as per normal setting during recalibration: if any of the two values exceeds the tolerances, the relevant error is displayed, the result of OD is shown in red and the previous valid values is kept as reference. The setting must be* ***B+C*** *to perform first the blank and then the calibrant.*
- *It is recommended to keep record of the calibration and reagent blank values, to check the change of diluent water and reagents; for this, use as a guideline the appendix Table 1. Sudden changes from the historical values must be checked for as soon as they appear, testing the followings:*

1. *the calibrant solution*
2. *the reagents solutions*
3. *the calibrant aspiration*
4. *the reagents aspiration*
5. *the dilution water change*

- *Before wasting an old set of working reagents and standards, it may be useful to use them to trace a malfunctioning within a new set, by cross checking the new and the old ones.*
- *The full scale of the system represents the maximum linearity of the analyser, and above this value the system will not go into automatic dilution. The dilution is not yet operative for TN measurement, while the TP analyzer is normally linear for all the normal water application (tested up to 1.5 ppm)*
- *When replacing the reagents it is advisable not to refill the actual bottle but to wash it and then refill with the fresh reagent, this to avoid carry over of old expired reagents*

Calibration and recalibration of the system

To perform the calibration of the system it is necessary to have a calibrant solution (see reagent preparation further on). The water and the wastes are placed outside the system, while the ***calibrant*** *and the* ***reagents*** *are left inside the system and follow the connections below:*

Straw name	*Description*
R1 AC	*Acid at 10 %*
R2 OX	*Oxidant solution*
R3 Colour P	*Colour reagent for PO4*
R4 Ascorbic	*Ascorbic acid for PO4*
C	*Calibrant solution at 1 ppm P-PO4 (may vary see final test)*
H	*DI water*
S	*Sample line*

The DI water is connected to the H labelled tube, and it is used for dilution, for blank measurement and the final wash at the end of the analytical cycles.

The calibration of the system is needed for one of the following reasons:

1. *the change of the reagents/calibrant solutions*

2. *the change or refill of the diluent water*
3. *the change of any of the hydraulic components*
4. *24 hours have passed after the last calibration*
5. *the 10 % tolerance on a grab known sample is exceeded.*
6. *the system has undergone a maintenance*

The description of the standard preparation is fully reported on the reagents preparation method, further on in this manual.

TABLE 1: Typical calibration OD values

Solution	***Cycle***	***OD***	***Note***
DI water	*Blank TP*	*0.001*	
3 ppm P	*Calibrant TP*	*0.50-0.6*	*Sample diluted inside digester*

The above values have been obtained at the Systea laboratory during the test of the system 2474. Please refer to the final test for a detailed system performance.

TABLE 2: Reagent consumption table per specific cycle: TP

Reagent or solution	***mL/analysis***	***mL/calibr.***	***mL/blank***	***mL/wash***
Acid 10 %	*0.36*	*0.36*	*0.36*	*-*
Oxidant	*0.36*	*0.36*	*0.36*	*-*
Colour	*0.36*	*0.36*	*0.36*	*-*
Ascorbic	*0.18*	*0.18*	*0.18*	*-*
Calibrant	*-*	*15*	*-*	*-*
Dilution water	*45*	*45*	*60*	*45*
Sample (factory timings)	*15*	*-*	*-*	*-*

For the reagent preparation, please prepare the reagents accordingly to the estimated consumption rate, considering the sampling frequency and the table above.
When replacing the reagents, please ensure a perfect cleansing of the reagent and calibrant bottles to avoid any expired reagent passing into the fresh reagents.
The sample aspiration rate is normally around 30 mL/min with the pump in FAST; the ***sampling time*** *with the pump in fast has to be set during the installation to allow a complete significant sampling, relatively to the length of the sampling line. This operation is normally performed by the technician during the installation. Please note that the sampling time in fast is not affecting the consumption of the reagents and of the dilution water. For touch screen systems this time is settable at the Timing Parameters menu in the first line, "SAMPLING TIME".*

Colorimeter calibration and balancing

The colorimeter for TP is a double beam single channel colorimeter, and the reading cell is the digester itself; removing the black aluminium front cover of the digester, it is possible to see the flowcell and the main components of the device, as shown in the picture. Some details are described in the box next to pictures. The light of the LED is not visible being at 880 nm.

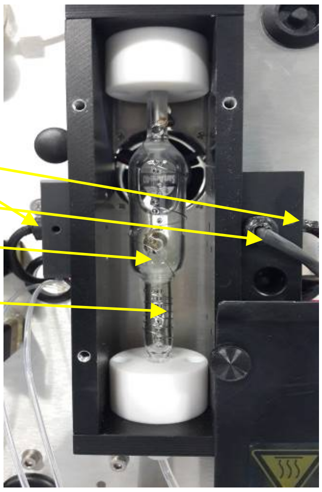
SAMPLE detector (through flowcell)

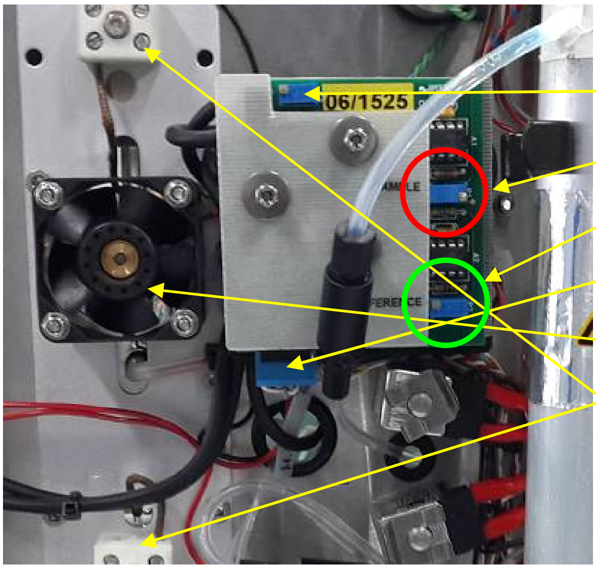
LED emitter 880 nm

REFERENCE detector

Digester – flowcell

Heating wire





LED 880 power trimmer (set at 13 mA)

SAMPLE amplifier trimmer

REFERENCE amplifier trimmer

Colorimeter connector

Cooling fan

Heater wire connections

Quick TP Colorimeter calibration - rebalancing.

With the flowcell filled with **DI water**, go to [Diagnostic] page and check the Color 1 sample and reference energies for touch screen systems, or under General Settings and Read Color. mV page for Graph display with keypad:

- Adjust the sample detector energy at 3.8 V acting on the SAMPLE trimmer shown above in red circle
- Adjust the reference detector energy at 4.0 V acting on the REFERENCE trimmer shown above in green circle

With these settings the final OD in flowcell will be 0.023 ± 0.003

If the above values can't be reached, a general setting of the energy can be done by increasing or decreasing the current to the LED by acting on the LED power trimmer, this will increase or decrease both the sample and reference detectors; furthermore, the LED positioning can be changed to modify the energy: this change is for specialist, not recommended for periodical check of the colorimeter calibration.

Troubleshooting guide

As a general rule, before looking for any probable cause of malfunctioning, check the following:

- *Power supply: 12 VDC (from 11.8 to 13.8 VDC, minimum 3 A); Heating bath 17 VDC*
- *Hydraulic connection: check the connection of tubes with the flowchart at the end of the manual*
- *Valve tightness: check the tightness of all hydraulic connectors, search for leaks*
- *Electrical connection: check visually the connection of all the wires in the rear of the manifold, and the insertion of the connector inside the electronic compartment (technician support may be needed, because the system may differ from standard)*
- *I/O devices: refer to the chapter related (**Manual I/O check procedure)** to test singularly any device present in the circuit*

In the picture below the back of the manifold is shown: the colorimeter board, the pump motor, the valves solenoids, the input board for sensors. The valves and motors connections are similar to normal Micromac layout, refer to service manual for wiring colour code.

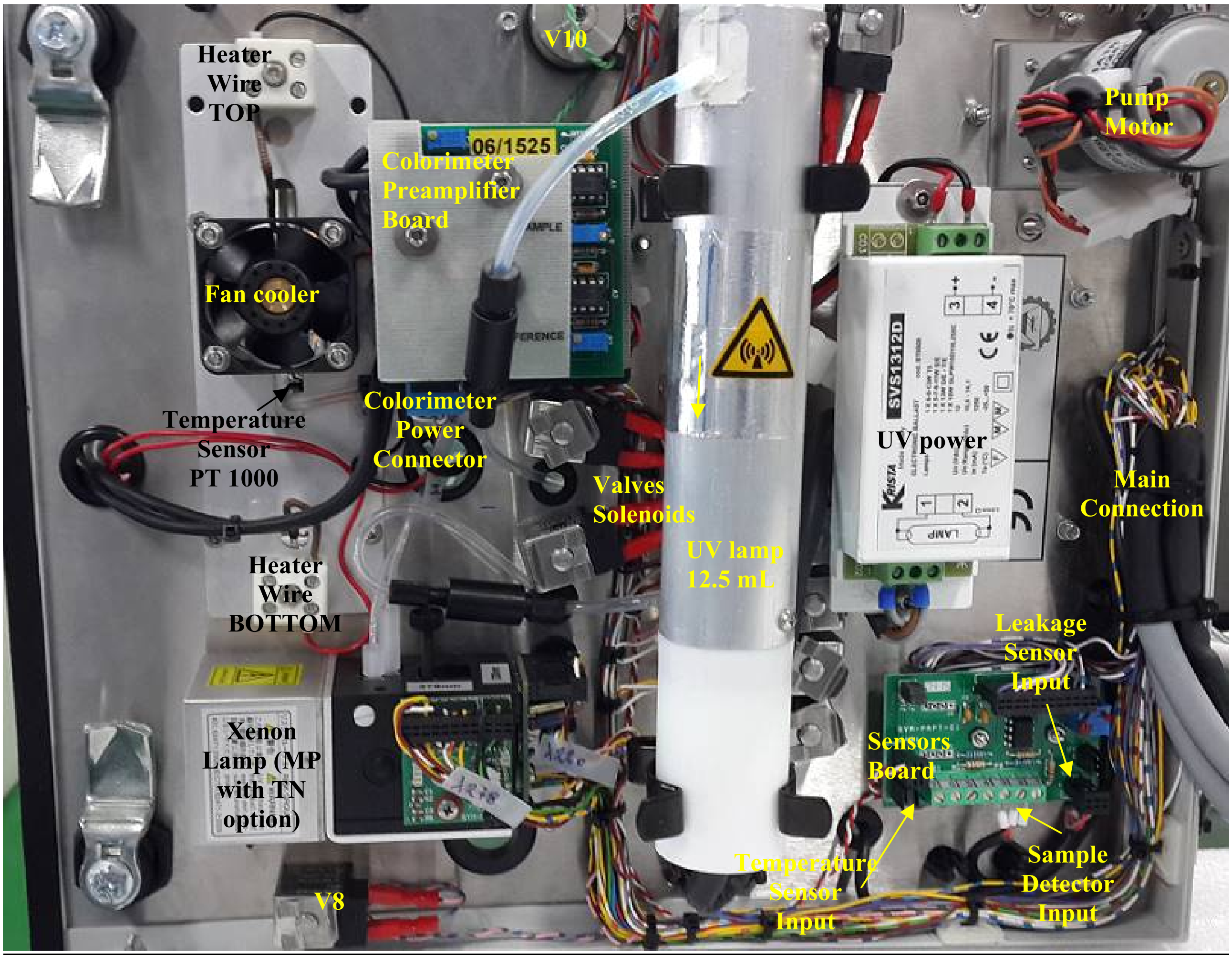


TABLE 3: Troubleshooting guide hints

Malfunctioning	*Probable cause*	*Remedy*
Missing result (with a known sample or standard)	▪ *Sample not aspirated*	▪ *Blockage of the sampling line: unblock or change the tube* ▪ *Breakage of the Pump (expected life:10000 hours): replace the tube* ▪ *Malfunctioning of the sample valve:* *1. Clean the valve* *2. Test power supply: must be within 11.8 and 13.8 VDC* *3. Check the valves connection wires* *4. Test valve "off line"* ▪ *Malfunctioning of the I/O driver board: replace the board.*
Missing result (with a known standard or sample)	▪ *Standard or known sample not valid* ▪ *Colorimeter not working*	▪ *Prepare standard again* ▪ *Prepare reagents again* ▪ *Check the colorimeter settings as described under "Quick colorimeter calibration - rebalancing"; open the digester front cover and check that LED is powered; disconnect and reconnect the colorimeter blue plug*
Result drift (values slowly changing in one direction)	▪ *Reagents close to expiry* ▪ *Temperature of digestion not stable* ▪ *Pump tube next to the expiry* ▪ *Colorimeter LED not stable*	▪ *Replace reagents* ▪ *Check temperature during digestion; check connections of heater; check PT1000 sensor is fully inserted* ▪ *Check pump flowrate; replace pump tube* ▪ *Recalibrate the colorimeter in DI water, check LED current to be at 13 mA, eventually regulate with LED trimmer and rebalance the colorimeter; (replace LED - replace photodiodes: service person)*
Response always higher than expected, on sample and calibrants	▪ *Reagents polluted* ▪ *Dilution water polluted* ▪ *Reagents polluted*	▪ *Replace reagents with fresh solutions, clean the reagent bottles during refilling* ▪ *Change dilution water with TN and TP free water, do not wash with soap the glassware* ▪ *Change reagents*

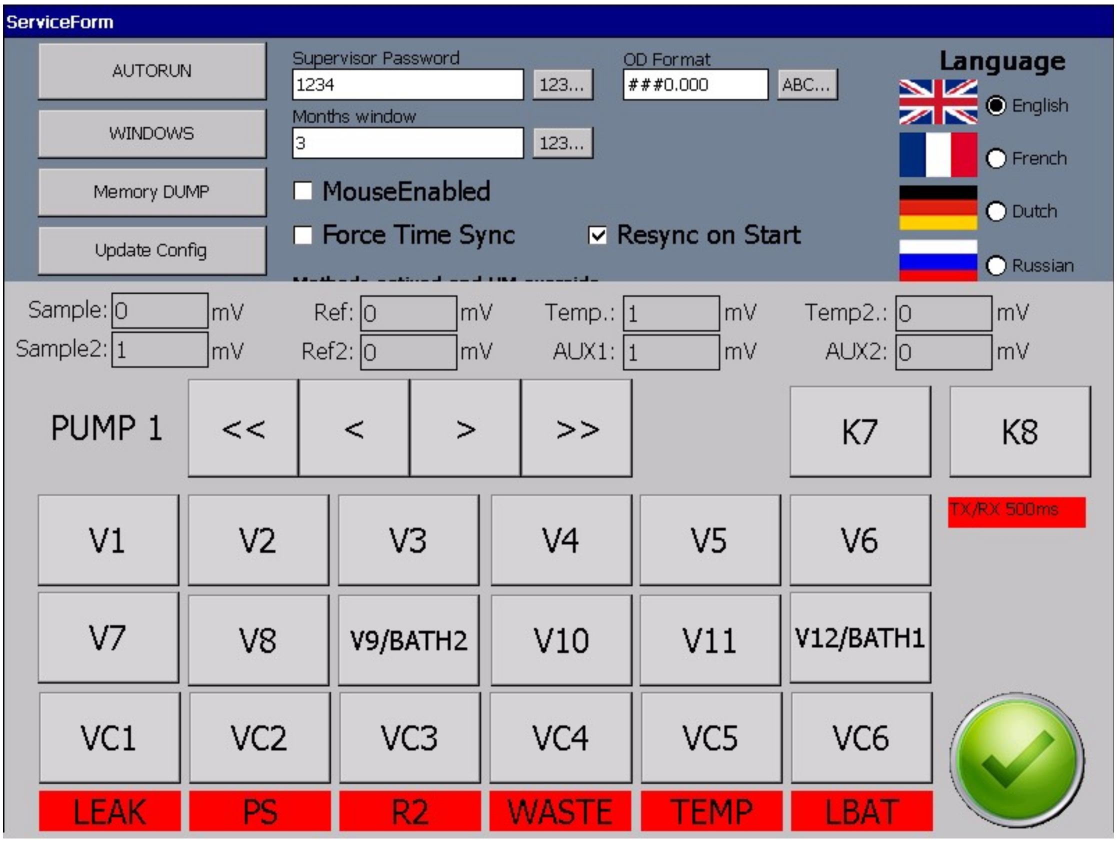
Malfunctioning	***Probable cause***	***Remedy***
Response drift	▪ *Reagents close to the expiry date.* ▪ *Pump Tube close to the expiry*	▪ *Change reagents and calibrant according to the method* ▪ *Change the pump tube*
Tube jumping during aspiration (pump on)	▪ *Waste line blocked or bent* ▪ *V8 blocked or obstructed* ▪ *Loop hydraulic obstructed, digester clogged* ▪ *Circuit tubes bent or obstructed*	▪ *Check the waste line and unblock, make the output as free as possible.* ▪ *Test the voltage power supply* ▪ *Test the valve wirings* ▪ *Test the valve with the "**Manual valve check procedure**" described in this manual, replace if necessary.* ▪ *Check the valve: it must be **open** towards the waste when not powered* ▪ *Check the circuit and clean with the aid of a syringe in counter flush direction (from outlet to inlet).* ▪ *Clean or replace the part blocked, with proper length of silicon tube 2x4 mm or Norprene tubing or Teflon 1.6x3.2 mm*
Response under the expected value	▪ *Reagents close to the expiry* ▪ *Calibrant close to the expiry*	▪ *Replace the reagents* ▪ *Replace the calibrant; replace the dilution water; check the colorimeter balancing*
Colorimeter start OD drifting	▪ *Colorimeter flowcell dirtying* ▪ *Colorimeter LED unstable*	▪ *Wash the hydraulic with NaOH 1 M aspirated as DI water (H) and run 2 washes with this solution: repeat the washes with DI water and check again the OD* ▪ *Check colorimeter in DI water and run system continuously in DI water: if drift still exists replace LED and rebalance colorimeter*
Response correct but with longer time of analysis	▪ *Cooling fan not working*	▪ *Check the fan cooler by Diagnostic (see further in this addendum) as V9* ▪ *Replace fan cooler*
Temperature not correct during analysis or stand by	▪ *Temperature sensor PT1000 not working*	▪ *Check temperature in Diagnostic page, must be around 500 mV (50°C) in standby* ▪ *Replace temperature sensor*

Manual valve and pump check procedure

The following procedure describes how to check any of the outputs of the µMAC, such as valves and the pump, controlling all the factors, from the Touch Screen, the CPU, passing through the I/O board, the wires to get up to the valve and pump. This test applies for the µMAC series with the graphic display as well as for the systems with touch screen.
The description of system with only touch screen is given under ***Test for graphic display and keypad****; the following steps describe how to run the test when a touch screen is installed.*

Test for touch screen

1) *Be sure the system is in standby (no monitor)*
2) *Go to the SERVICE page pressing the button on the right side (at this time a password is required, it is 3147)*
3) *Enter the diagnostic page pressing the Diagnostic button*
4) *A page as shown below will appear*



5) *to activate or deactivate any of the component of the manifold, simply press ONCE the relevant button and wait for the activation (buttons dark grey): being the communication serial it may take some time to start or stop the output; the pump buttons meaning are, from left to right:[<<] pump fast reverse, [<] pump slow reverse, [>] pump slow forward, [>>] pump fast forward*

6) *The activation or deactivation of the outputs may be multiple, but great attention has to be put on multiple activation because this may cause serious consequences on the system*
7) *In this page the colorimeter photodiodes energies are also displayed, they are called Sample: and Ref: simply, and the boxes show the energies in mV: set the energies as described in "Quick Colorimeter calibration – rebalancing" for further details*
8) *The temperature output here is represented in mV, where 10 mV=1°C, so a display of 275 mV means a temperature of 27.5°C*
9) *When leaving this page by the green button, all the output MUST be set to off (light grey colour) or the following cycles may not work properly*

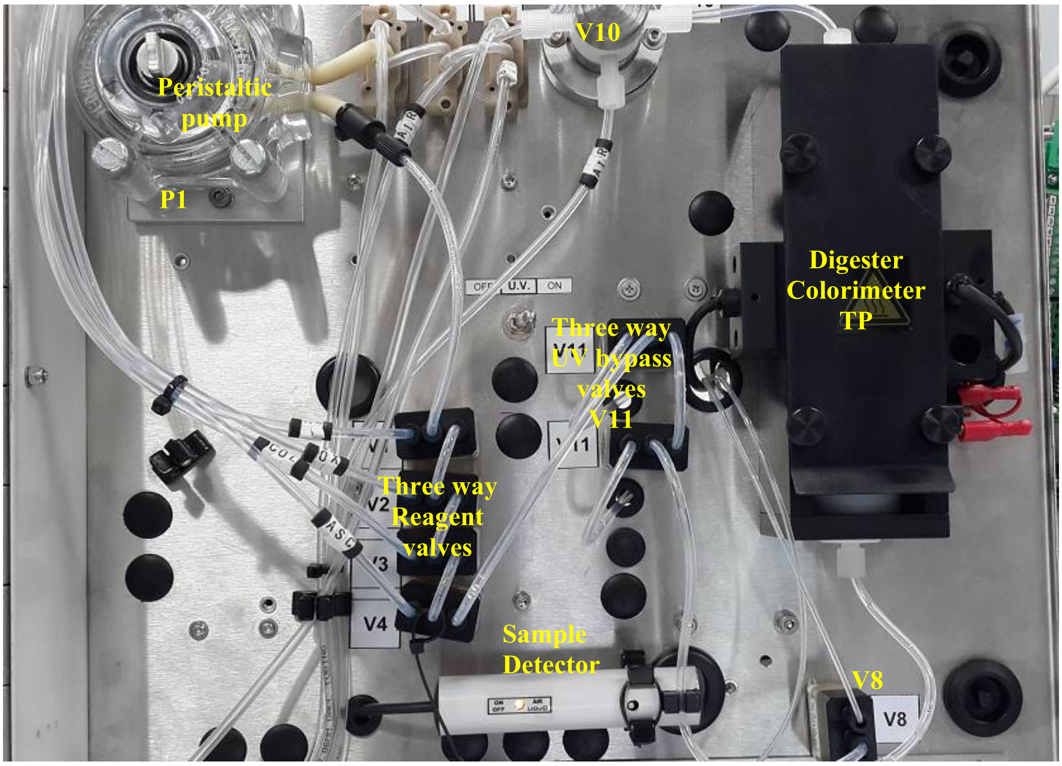
Note: do not start any VC command for TP-TN system, being all output used for Xenon lamp.

Test for graphic display and keypad

To perform the test, the only things needed are:

1. *The flowchart of the system to locate the valves*
2. *The list of the command to activate or deactivate the valves (available from Systea), also drawn on the flowchart for the main valves*
3. *The system switched on and in standby mode, with a stable power supply of **12** VDC(11.8-13.8).*
4. *the keypad connected and the system rebooted (the keypad is only started at system start up)*

***Photo**: the picture represents the COD Micromac C from **front view**, and the consideration may be extended to any device, considering that the valves are three way.*



*As first step of the check, the valves may be tested manually, even without any power to the system, just connecting a syringe to the common **C** or normally open **NO** part of the valve, and check the free movement of liquid between these two channels: use DI water for this check.*
*For complete testing of the valve, attach the syringe to the common **C** output, then test alternatively both the normally open **NO** and the normally closed **NC**, activating the valve as described in following section.*

To perform the complete check of any of the output, follow the procedure described below:

1. *Check the power supply of the system in stand by, measuring the input voltage: it must be 12.5 VDC (11.8 to 13.8); the low voltage is in fact the first reason for the valve not to be excited, especially during multiple valve activation. If the voltage falls below 11 VDC, some valve may be deactivated without any notice.*
2. *On the keypad, press [F] to show the main menu.*
3. *Select [Parameter] then [E] (enter).*
4. *Select [Cycle editor] then [E] (in case of multiparametric system, any method is valid).*
5. *Select the [New Cycle] (on top of the list) then [E]. The system will now show a grid with the left hand column enumerated from 0 (zero) on. The cursor will be on the first free position on the top left part of the screen, under the line Command.*
6. *With the help of the flowchart, trace down the valve to be checked and the commands related to its activation and deactivation. If the flowchart just reports the valve number, find out the codes in the list of the command for the µMAC series instruments (provided by Systea).*
7. *Type in the activation code for the valve, then press [E]: the cursor will move to the right, under the column Time, type here the seconds for the activation of the valve, then press [E]. The cursor will automatically go on the following step (the third column, under Tick, is ignored from most of the valve, and is requested automatically from the system only for the valves using this setting).*
8. *Type here the deactivation code for the valve, then press [E], and type the time for this command, normally 2 seconds, confirming it by [E]. It is necessary here to advice that the minimum time for any command to be operative is one second.*
9. *Press [F]to go to the cycle editor submenu, , scroll up to the [Test sequence] and press [E]: the selected valve will be switched on for the set time, then will be deactivated. Repeat the [Test sequence] to check the valve again, if necessary.*
10. *Pressing the left arrow key [<-], the screen will go back to the command list, and other devices may be checked if needed. A multiple check is also possible, it is only necessary to type in all the commands of the valves that have to be tested.*
11. *Once terminated this procedure, to get out from this list press [F] to access at the cycle editor submenu, then scroll down to [Exit] and press [E].*
12. *The system doesn't remember the sequence tested unless it is stored, but sometimes some command codes may still be "on", even after the exit from the cycle editor, for this it is recommended to go to the main menu and press <Stop>, to reset any command definitely.*

Example of Test procedure using Keypad and graphic display

*To clarify better this procedure, let's make an example with some valves that are present in the system. The valves to be checked be the **V7** (three way water valve) and the **V5** (three way sample valve).*
The command for the activation and deactivation of these valves are:

V7 : 122** = valves **ON**   **123** = valves **OFF

***V5 : 118** = valve **ON**   **119** = valve **OFF** .*

Procedure test:

1. *Press [F], then scroll up to [Parameter] and press [E]*
2. *Scroll to [Cycle editor] and press [E] (for the MP, any method is suitable)*
3. *Scroll to [New cycle] and press [E]*
4. *The display will show a blank sequence, starting from step 0, with the cursor blinking on the first free row, under the "Command" message.*
5. *Type [**122**] then press [E](code for activation of V7), the cursor will move on the right, type [**2**] then press [E](seconds of activation of V7), the cursor will move to the next step.*
6. *Type [**123**] then press [E](code for reset of V7), the cursor will move on the right, type [**2**] then press [E](seconds of reset of V7), the cursor will move on the next step.*
7. *The cursor is now on the third line, type [**118**] then press [E](command for activation of V5), the cursor will move on the right, type [**2**] then press [E](seconds of activation of V5), the cursor will move to the next step.*
8. *Type [**119**] then press [E](command for the reset of V5), the cursor will move right, type [**2**] then press [E](seconds for reset of V5), the cursor will move to the next line.*
9. *Type [**F**] to exit from the cycle and enter into the cycle editor submenu. Scroll up to [Test sequence] and press [E].*
10. *The system will now perform the test cycle, activating the V7 valve for 2 seconds, then resetting the V7 for 2 seconds, then will activate the V5 for 2 seconds, and finally will reset the V5. The sequence may be repeated for any need, just repeating the [Test sequence]*
11. *The observation of the valve behaviour using a syringe will drive the check , and the noise produced by the valve activation, will mean that the whole test has been all right. To activate the valve ON for a longer period, e.g. to check the hydraulic by the aid of a syringe, just set a longer activation time instead of the 2 seconds of the example given.*
12. *After the test has been done, remind to exit from the cycle editor submenu by [exit], then reset any command in the μMAC memory by the <Stop> in the main menu.*

Scheduled maintenance

To keep the system running properly, a scheduled maintenance is required. Follow the table below and eventually reprint and leave close to the system. Missing the schedule may produce malfunctioning on the system, or decrease its performances.

TABLE 5: scheduled maintenance guide

OPERATION	***FREQUENCY***				
	daily	*weekly*	*monthly*	*Quarterly*	*annual*
Visual check of ERROR alarm indicator	*				
Visual check of liquids enclosures for leakages detection	*				
Visual check of reagents presence in containers	*				
Visual check of colorimeter LED working		*			
Check of cooler fan functioning		*			
Check of the tightening of the pump screws			*		
Calibration check against historical values		*			
Waste line and container check		*			
Sample line cleaning (depending on matrix)		*			
Visual check of the hydraulic manifold			*		
Pump tubing replacement					*
Digester check for leakages (for qualified personnel only)				*	
Hydraulic sleevings replace (for qualified personnel only)					*
Check of the colorimeter energies (for qualified personnel only)				*	
Analyzer general inspection (for qualified personnel only)				*	
Check of temperature sensor functioning			*		

TABLE 6: calibration historical values

Write here the calibration values obtained just below the reference ones, to keep a record.

Date	**Blank OD**	**CALIBRANT**	**Note**
26th June,2019	*0.004 OD*	*3 ppm 0.593 OD*	*S.7”+ H_2O.27” +H.2”+OX.2”+Col.2”+Asc.1”*

The first lines values have been obtained in Systea during the test of the system, the note means the timing of each component of the final solution, Sample water acid oxidant colour and ascorbic.

Total phosphorous in water

0-30 mg/L up to 40 ppm P-PO4

(uLFRHT-Ptot-01 rev.1 comm.3155)

Ref. U.S. Environmental Protection Agency - Method 365.1 - Determination of Phosphorus by semi-automated colorimetry.

Method description for Total Phosphorous

The sample is acidified and heated to hydrolyze all the inorganic complex forms of phosphorous, then it is oxidized by the double action of potassium persulfate and UV radiation in acid environment. The phosphate produced reacts in acid solution with molybdate to form phosphomolybdate, which is reduced to molybdenum blue by ascorbic acid.
The complex is read at 880 nm. Antimonium is used to increase the sensitivity.

Reagent preparation

N.B.: where not specified , all chemicals should be of A.C.S. grade. Water refers to high quality reagent water, as described in ASTM, type I or II.

- The preparation of the reagents requires a qualified person, the wearing of safety glasses, chemical resisting gloves, safety clothes protection and operation in adequate environment.

- Method interferences can be caused by improper reagent preparation, or contaminants in the reagents, reagent water, glassware, etc., which may bias the results. Take care to keep all such items free of contaminants.

- **The toxicity or carcinogenicity of each compound or reagent used in this method has not been fully established. Treat each chemical as a potential health hazard. Reduce exposure to these chemicals to the lowest possible level.**

- The following chemicals used in this method may be highly toxic or hazardous. Handle with extreme caution at all times. For reference purposes, make available the Material Safety Data Sheets (MSDS) for each chemical used in this method to all personnel involved in this chemical analysis.

List of raw materials

Description of the products	Chemical formula	Code	Safety classification
Ascorbic acid	**$C6H8O6$**	**VWR 20150.290**	irritation to the skin
Ammonium molybdate	**$(NH_4)Mo_7O_{24}$ *$4H_2O$**	**Merck 1.01182.0250**	danger, toxic
Antimony potassium tartrate	**$K(SbO)C_4H_4O_6$ *1/$2H_2O$**	**Merck 1.08092.0250**	toxic
Potassium dihydrogen phosphate	**KH_2PO_4**	**Merck 1.04873.1000**	---
Potassium persulfate	**$K_2S_2O_8$**	**Merck 1.05091.1000**	harmful, oxidizing
Sulfuric acid, **conc.96-98%**	**H_2SO_4**	**Merck 1.00731.1000**	corrosive
Riboflavine-5'phosphate salt	**$C_{17}H_{20}N_4NaO_9P$**	**Fluka 83810**	---
Tetrasodium diphosphate, x $10H_2O$	**$Na_4P_2O_7$*$10H_2O$**	**Merck 1.06591.0500**	---
Chloroform	**$CHCl_3$**	**Merck 1.07024.1000**	toxic

R1 Sulphuric acid H_2SO_4 10%

- Sulfuric acid **conc. 96-98% H_2SO_4 (Merck 1.00731.1000)** 50 mL
- DI Water q.s.500 mL

Preparation: Work under fume hood! In a 500 mL volumetric flask with about 250 mL of DI water pipette cautiously and while cooling 50 mL of **H_2SO_4 conc. 96-98%** let cool down then bring to volume with DI water and homogenize. Transfer in a plastic bottle, keep well closed. Stable for **four months.**

R2 Oxidant 4%

- Potassium persulfate **$K_2S_2O_8$ (Merck 1.05091.1000)** 20 g
- DI Water q.s.500 mL

Preparation: In a 500 mL volumetric flask, containing 250 mL of DI Water, transfer 20 g of potassium persulfate, mix until the salt is completely dissolved, bring to volume with DI water and homogenize. Transfer in a plastic bottle, stable for **one month,** when refrigerated at +4°C.

R3 Colour

- Antimonyl potassium tartrate **K(SbO)C4H4O6 x0.5H2O (Merck 1.08092.0250)** 0.24 g
- Sodium molybdate dihydrate **NaMoO4 x2H2O (Merck 1.06521.0250)** 11.1 g
- Sulfuric acid, **conc. H2SO4 (Merck 1.00731.1000)** 54 mL
- DI Water q.s.500 mL

Preparation: **Work under fume hood!** In a 500 mL volumetric flask containing about 200 mL of DI water, pipette cautiously while cooling and stirring 54 mL sulfuric acid conc.; in another flask containing about 100 mL of DI water dissolve the two salts, mix until complete dissolution. Then add very slowly the salts solution to the cool acid solution. Bring to volume with DI water and homogenize. The solution should be colorless. Transfer in a plastic bottle. Stable about **four months**, when refrigerated at +4°C. Transfer in a plastic bottle.

R4 Reducing agent 10%

- Ascorbic acid **$C_6H_8O_6$ (VWR 20150.290)**	50 g
- DI Water	q.s.500 mL

Preparation: In a 500 mL volumetric flask containing about 250 mL of DI water dissolve 50 g ascorbic acid, mix until complete dissolution, bring to volume with DI water and homogenize. Transfer in a plastic (**dark**) bottle, and keep cool when not in use. Stable at least **one month.**

Standard stock P at 1000 mg/L PO4 as P

- Potassium hydrogen phosphate anhydrous **KH_2PO_4 (Merck 1.04873.1000)**	4.394 g
- Chloroform $CHCl_3$ **(Merck 1.02445.1000)**	10 drops
- DI Water	q.s.1000 mL

Preparation: In a 1000 mL volumetric flask containing about 400 mL of DI water, transfer exactly 4.394 g of KH_2PO_4; mix until complete solution. Bring to volume with DI water and homogenize. Add 10 drops of chloroform as preservative, transfer in a dark glass bottle, stable **six months,** refrigerated at +4°C.

Working calibrant 30 mg/L P-PO4

- Stock standard P at 1000 mg/L P-PO4	30 mL
- DI Water	q.s.1000 mL

Preparation: In a 1000 mL volumetric flask containing about 800 mL of DI water transfer exactly 30 mL of standard stock P 1000 mg/L P-PO4. Bring to volume with DI water and homogenize. Use this solution for the periodical recalibration of the system.

Standard stock 100 mg/L Ptot (organic) as P for the UV lamp check

The recovery test is performed with the aid of the riboflavine-5’ phosphate, that is a organic phosphorous compound.

- Riboflavine-5’ phosphate monosodium salt **$C_{17}H_{20}N_4NaO_9P$ (Fluka 83810)**	0.1544 g
- DI Water	q.s.100 mL

Preparation: In a 100 mL volumetric flask with about 80 mL of DI water dissolve the salt Dissolve the salt in about 80 mL of DI water, mix until complete solution. Bring to volume with DI water and homogenize. Transfer in a plastic bottle. Stable about **six months**. This solution appears to be highly coloured in yellow.

<u>Standard stock A 1000 mg/L P tot (inorganic) as P for the high temperature digestion check</u>

The recovery test is performed with the aid of the $Na_4P_2O_7$ x10 H_20, tetrasodium diphosphate decahydrate, that is a inorganic phosphorous compound.

- Tetrasodium diphosphate decahydrate **$Na_4P_2O_7$ *10 H_20 (Merck 1.06591.0500)** 1.8 g
- DI Water q.s.250 mL

Preparation: In a 250 mL volumetric flask with about 150 mL of DI water dissolve the salt, mix until complete solution. Bring to volume with DI water and homogenize. Transfer in a plastic bottle. Stable at least **six months**. This solution appears to be colourless.

Diluent water

It is used both for reagent blank and washings. Use high quality reagent water when run a reagent blank cycle and standard DI water during normal run.

<u>Method revision</u>

Rev.	Description
0	First release
1	**Hazard symbols changed 17/01/2020**

Note: For best performances, the DI Water and calibrant should be placed at the level of the analyzer.

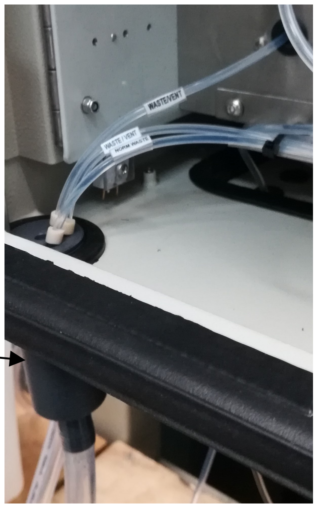
<u>Reagent consumption</u>

Reagent #	Description	Aspiration per test (sec.)	Volume per test (mL)	Volume per 1000 test (mL)
R1	**H2SO4 10% (3.6N)**	**2 sec.**	**About 0.4 mL**	**About 400 mL**
R2	**OX- 4%**	**2 sec.**	**About 0.4 mL**	**About 400 mL**
R3	**Colour**	**2 sec.**	**About 0.4 mL**	**About 400 mL**
R4	**Reducing 10%**	**2 sec**	**About 0.4 mL**	**About 400 mL**

Start up of the system

1) prepare all reagents as described in the method
2) place the reagent straws into their proper reagent bottles, respecting the reagent number
3) place the calibrant straw (label Cal) into the calibrant solution bottle
4) place the diluent water (label DIW) into the DI water container, normally a 5 liters bottle, that should be placed just below the system
5) place the waste line (label Waste) into a proper container, and dispose of it properly (check for the treatment of the wastes according to the local law enforcement)

Drain/waste receptacle from bottom of system



6) switch on the system
7) start a **[Prime]** cycle on the [Operation] menu, repeat twice
8) start a **[Reagent blank]** cycle on the [Operation] menu
9) check the OD of the blank (if out of tolerance respect to the Systea internal test, repeat the reagent blank: if it is still out of tolerance, check the reagent preparation and the quality of the water)
10) start a **[Calibration]** cycle on the [Operation] menu
11) check the OD of the calibrant (if out of tolerance respect to the Systea internal test, repeat the calibration: if it is still out of tolerance, check the reagent and standard preparation and the quality of the water)
12) start an **[Analysis]** cycle on the [Operation] menu using as **sample** a solution of 50 % of calibrant solution and 50 % of DI water, and check its value: it must be the 50 % of the calibrant value. This check will confirm the linearity of the system and its proper functioning
13) the system is now ready for the analysis.

The above procedure must be performed at any reagent change.
A record of the calibration and reagent blank values is recommended, to keep trace of the various calibration of the system.
This will produce a sequence of data that will help in tracing any trouble in case of reagents change.

Place in front of hydraulic door (outside) the following reagents:

R1 Sulphuric Acid 10%;
R3 Color;

Place inside the analyzer the following solutions:

R2 Oxidant 4%;
R4 Reducing reagent 10%.
The Calibrant.

DI Water can be placed under the analyzer.

Micromac TP cycle description comm. 2474, sample diluted inside digester

Brief description of the analytical sequences of the system: look at the flowchart for further information.

The calibration cycle and the blank cycle are the same of the analysis reported here, the change is related to the calibrant and DI water valves, V6 and V7: the location of the proper steps is highlighted in green; the amount of sample and dilution water are highlighted in light blue.

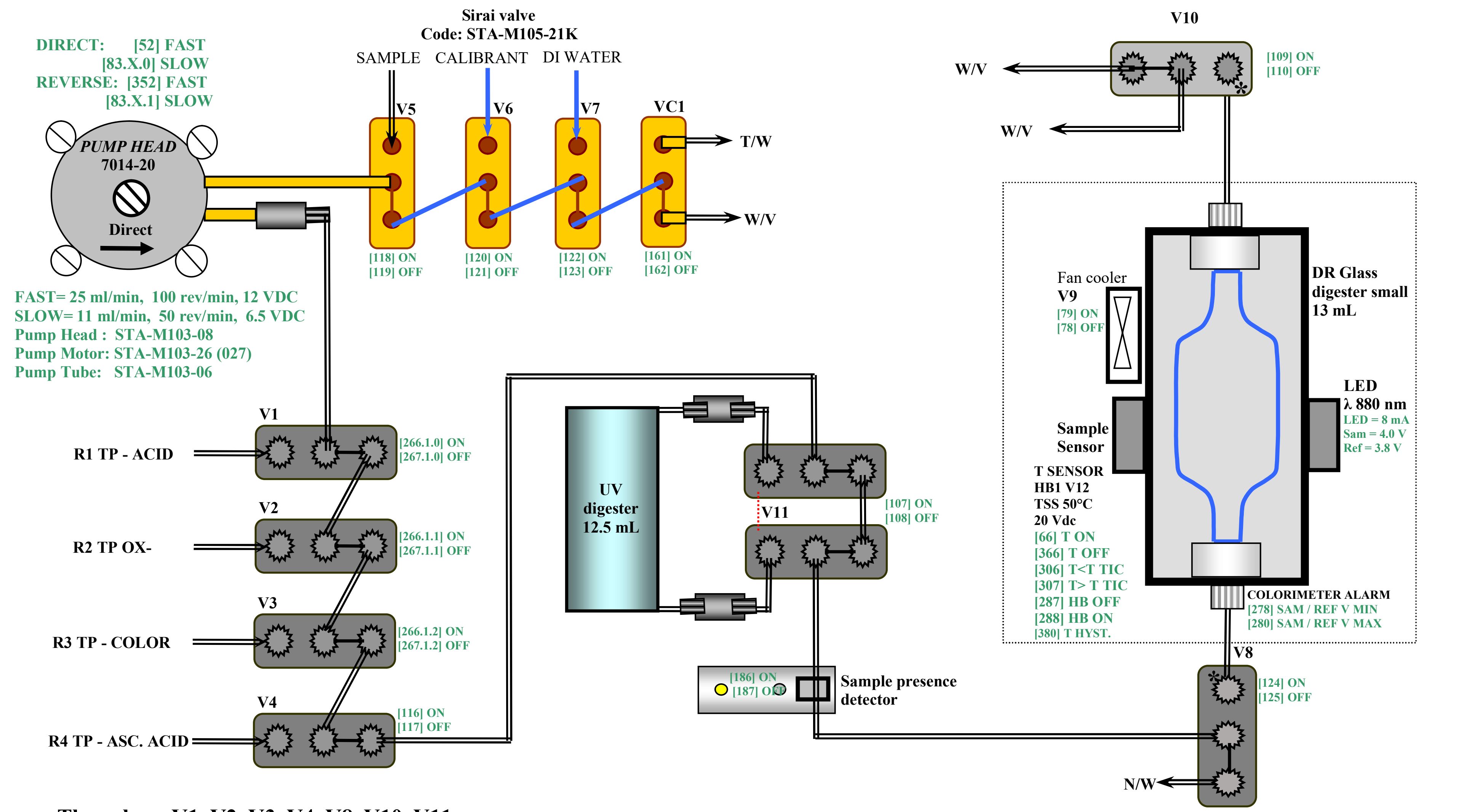
The above sequence is referred to the sample analysis, and it is the same used for calibrant and blank: for systems with touch screen the digestion time for both UV and heating bath are settable under Timing Parameters as "UV Digestion time" and "HB Digestion time", in seconds.

Notes

Micromac C TOT P spare part list with flowchart

P/N	*DESCRIPTION*	*ID*
STA-M105-21K	**3 WAY VALVE SIRAI BARBED (Kalrez)**	**V5, V6, V7**
STA-M105-27K	**3 WAY VALVE SIRAI (Kalrez)**	**V1, V2, V3, V4, V11**
STA-M106-06	**Block connector in PVC**	**V1, V2, V3, V4, V11**
STA-M105-27K	**3 WAY VALVE SIRAI (Kalrez)**	**V8, V10**
STA-M106-08	**Block connector in Teflon**	**V8, V10**
STA-NUT-04	**NUT in PEEK**	**V8, V10**
STA-L880-01F5	**EMITTER 880 nm, 5 mm dia**	**LED**
562-3019-X5	**CRYSTAL TUBE PFA-1.6x3.2 mm**	**Hydraulic circuit and reagents tube**
116-0536-14	**TYGON TUBE 1.85 mm ID**	**Connection tube Sample (S), Calibrant (C), DIW(H), Waste/ vent**
COD-DIGEST-06	**SMALL GLASS DIGESTER WITH RESISTOR**	**SMALL GLASS DIGESTER**
STA-UVLM-09	**UV DIGESTER LAMP (12.5 ml) Double coil**	**UV LAMP**
562-3019-X2/6	**NUT adapter UV Lamp**	**NUT adapter**
STA-M103-08	**PUMP HEAD 7014-20 MIC C**	**PUMP**
STA-M103-26	**MOTOR BRUSHLESS 12 VDC100 RPM Standard for MIC C**	**MOTOR**
STA-M103-06	**PUMP TUBE**	**PUMP**
STA-CONN-01	**NUT ADAPTER for PUMP TUBE**	**NUT ADAPTER**
STA-NUT-F01	**NUT BLACK for flanged crystal tube 562-3019-X5**	**NUT BLACK**
109P0412-G3023	**FAN COOLER**	**FAN COOLER**
COD-PTDR-00	**T° sensor HBI**	**T° sensor**
PRS-SNSR-01	**SAMPLE DETECTOR**	**SAMPLE DETECTOR**

SYSTEA



The valves: V1, V2, V3, V4, V8, V10, V11
SMC Code: STA-M105-16
Valve manifold in PVC code. STA-M106-03
Valve manifold in Teflon code: STA-M106-09 for the valves: V8, V10

Legend: **Place the reagents:**
R1, R3 in front of hydraulic door (outside).
R2, R4 and Calibrant inside the analyzer;